

NATIONAL UNIVERSITY OF SINGAPORE
Department of Mathematics

MA1513
Additional Practice Set 4

1. (Do not use MATLAB for this question) Let $\mathbf{A} = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 1 & 1 \\ 0 & 1 & 0 \end{pmatrix}$.
- (i) Find the eigenvalues and corresponding eigenvectors of $\mathbf{A}$. Show your working.
 - (ii) Is $\mathbf{A}$ diagonalizable? Why?
 - (iii) How many solutions does the homogeneous system $\mathbf{A}^{100}\mathbf{x} = \mathbf{0}$ have? Justify your answer.

2. (Do not use MATLAB for this question) Given matrix $\mathbf{A} = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 2 & 2 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 4 \end{pmatrix}$.

- (i) Write down the eigenvalues of $\mathbf{A}$.
- (ii) Is $\mathbf{A}$ diagonalizable? Why?
- (iii) Which of the following are eigenvectors of $\mathbf{A}$?

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \end{pmatrix}, \mathbf{v}_3 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \mathbf{v}_4 = \begin{pmatrix} 3 \\ 4 \\ 2 \\ 0 \end{pmatrix}, \mathbf{v}_5 = \begin{pmatrix} 8 \\ 12 \\ 9 \\ 3 \end{pmatrix}.$$

- (iv) Find a matrix $\mathbf{P}$ such that $\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ is a diagonal matrix.

3. Let $\mathbf{A}$ be the matrix $\begin{pmatrix} 2 & -1 & 1 \\ 1 & 0 & 1 \\ 1 & -1 & 2 \end{pmatrix}$.

- (i) Determine whether $\mathbf{u} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$, $\mathbf{v} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$, $\mathbf{w} = \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix}$, $\mathbf{x} = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$ are eigenvectors of $\mathbf{A}$.
- (ii) Use (i) to write down the characteristic polynomial of $\mathbf{A}$.
- (iii) Without using MATLAB, write down a basis for each eigenspace of $\mathbf{A}$.
- (iv) Without using MATLAB, write down a non-singular matrix $\mathbf{P}$ and a diagonal matrix $\mathbf{D}$ such that $\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \mathbf{D}$.
- (v) Write down the eigenvalues of $4\mathbf{A}^{-1}$.

4. A 3×3 matrix $\mathbf{A}$ has characteristic polynomial $x^3 - 2x^2 + x$ and two eigenspaces given by

$$\text{span} \left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right\} \quad \text{and} \quad \text{span} \left\{ \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\}.$$

- (i) Find a 3×3 non-singular matrix $\mathbf{P}$ and a diagonal matrix $\mathbf{D}$ such that $\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \mathbf{D}$.
- (ii) Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $T(\mathbf{x}) = \mathbf{A}\mathbf{x}$. Find the subspace V (in terms of a linear span) of $\mathbb{R}^3$ that consists of all the points that are mapped to the zero vector $\mathbf{0}$ under the linear transformation T .
- (iii) A certain species of animals can be classified as babies, cubs and adults. There are 400 babies, 300 cubs and 100 adults initially. Let a_n, b_n and c_n be the number of babies, cubs and adults respectively after n years. A population model is given by

$$\begin{pmatrix} a_n \\ b_n \\ c_n \end{pmatrix} = \mathbf{A} \begin{pmatrix} a_{n-1} \\ b_{n-1} \\ c_{n-1} \end{pmatrix}$$

where $\mathbf{A}$ is the matrix above.

Explain why the population of the three groups of the animals remain the same every year based on this model.

5. $\mathbf{M}$ is a 3×3 matrix such that

$$\mathbf{M} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{M} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 4 \end{pmatrix}, \quad \mathbf{M} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix}.$$

- (i) Without finding $\mathbf{M}$, determine $\mathbf{M} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ and $\mathbf{M} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$.
 - (ii) Use the given information and part (i) to find $\mathbf{M}$.
 - (iii) Without finding $\mathbf{M}^{-1}$, determine the 3×3 matrix $\mathbf{X}$ such that $\mathbf{M}\mathbf{X} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$.
 - (iv) Use part (iii) to find $\mathbf{M}^{-1}$.
6. $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is a linear transformation which has the effect of rotating about the z -axis anticlockwise by 90° , followed by the reflection about the xy -plane.
- (i) Find the images of $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ under T .
 - (ii) Find the standard matrix $\mathbf{A}$ such that $T(\mathbf{x}) = \mathbf{A}\mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^3$.

- (iii) By multiplying the matrix $\mathbf{A}$ in (ii) to the complex vector $\begin{pmatrix} 1 \\ -i \\ 0 \end{pmatrix}$, show that this is an eigenvector of $\mathbf{A}$ and write down its corresponding (complex) eigenvalue.
7. Given a linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ such that it scales along the line with equation $y = x$ on the xy -plane by a scaling factor 2, and scales along the y -axis by a scaling factor $\frac{1}{2}$.
- Determine the standard matrix of T .
 - Sketch the image of the parallelogram with vertices $(2, 2), (0, 4), (-2, -2), (0, -4)$ under the transformation T .
 - Find the vector that is transformed to $\begin{pmatrix} 8 \\ -4 \end{pmatrix}$ under T .
 - Find the image of $\begin{pmatrix} 16 \\ 0 \end{pmatrix}$ by performing T consecutively for 3 times.
8. Consider a population model for a certain species of animals that has two life stages: juvenile (up to 1 year old) and adult. Each year,
- an adult produces an average of 1.6 juveniles;
 - 30% of the juveniles survive to become adults;
 - 80% of the adults survive.
- Let $\mathbf{x}_n = \begin{pmatrix} j_n \\ a_n \end{pmatrix}$ where j_n and a_n are the number of juveniles and adults respectively in year n .
- Find the stage matrix $\mathbf{A}$ such that $\mathbf{x}_{n+1} = \mathbf{A}\mathbf{x}_n$.
 - Find the inverse matrix $\mathbf{A}^{-1}$.
 - Suppose the number of juveniles and adults are 480 and 360 respectively in year 2019. Find $j_{2018}, a_{2018}, j_{2020}, a_{2020}$.
 - Compute the ratio $\frac{j_n}{a_n}$ for $n = 2018, 2019$ and 2020. What do you observe? Give an explanation of your observation.
9. The annual migration of a population between a city and its suburbs are given by the following model:
Each year, 10% of the city population moves to the suburbs, and 5% of the suburbs population moves to the city.
- Let a_n and b_n denote the population in the city and its suburbs respectively at the end of year n . Find the matrix $\mathbf{A}$ such that

$$\begin{pmatrix} a_{n+1} \\ b_{n+1} \end{pmatrix} = \mathbf{A} \begin{pmatrix} a_n \\ b_n \end{pmatrix}.$$

- (ii) Suppose the current population of the region is 600,000 in the city and 300,000 in the suburbs. Find the distribution of the population between the city and its suburbs in the long run.
10. On any given day, a student in School X is either healthy or sick. Of the students who are healthy today, 95% will be healthy tomorrow. Of the students who are sick today, 55% will still be sick tomorrow.

Let $\mathbf{x}_n = \begin{pmatrix} h_n \\ s_n \end{pmatrix}$ where h_n and s_n are the number of students who are healthy and sick respectively in day n .

- (i) Find the 2×2 matrix $\mathbf{A}$ such that $\mathbf{x}_n = \mathbf{A}\mathbf{x}_{n-1}$.
- (ii) Suppose 20% of the students in the school are sick on a particular day. What percentage of the students are likely to be sick on the following day?
- (iii) At the steady state, what is the proportion of students in School X who are sick on any given day?